Power Budget

Team Number: 209
Project Name: **Automated Laundry Folding Robot**
Team Member Names: **Diego, Ruben, Zainan, Kristen**
Version: **1**

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	DC Gearmotor	FIT0186	12 V	1	7,000.00	7,000.00	mA
	H-Bridge Motor Driver	DRV8876PWPR	4.5-37 V	1	7.00	7.00	mA
	PIC18F57Q43 Curiosity Nano	DM164150	1.8-5.5 V	1	100.00	100.00	mA
	Quad Op-Amp	MCP6004-I/P	1.8-6 V	1	0.68	0.68	mA
	Reflective IR Sensor	OPB732	5 V emitter circuit	1	50.00	50.00	mA
	Motor Hall Encoder	Integrated in FIT0186	5 V	1	20.00	20.00	mA

+12V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	DC Gearmotor	FIT0186	12 V	1	7,000.00	7,000.00	mA
	H-Bridge Motor Driver	DRV8876PWPR	4.5-37 V	1	7.00	7.00	mA
						0.00	mA
					Subtotal	7,007.00	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	8,758.75	mA
Source	12 V Wall Supply	GST60A12-P1J	12 V regulated output	1	5,000.00	5,000.00	mA
					Total Remaining Current Available on +12V Rail	-3,758.75	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity Nano	DM164150	1.8-5.5 V	1	100.00	100.00	mA
	Quad Op-Amp	MCP6004-I/P	1.8-6 V	1	0.68	0.68	mA
	Reflective IR Sensor + IR LED	OPB732	5 V emitter circuit	1	50.00	50.00	mA
					Subtotal	150.68	mA
					Safety Margin	25%	

					Total Current Required on +5V Rail	188.35	mA
Regulator	+5V Linear Regulator	L7805CV	7-35 V input	1	1,500.00	1,500.00	mA
					Total Remaining Current Available on +5V Rail	1,311.65	mA

Unused Power Rail	**Component Name**	**Part Number**	**Supply Voltage Range**	**Qty.**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
						0.00	mA
						0.00	mA
					Subtotal	0.00	mA
					Safety Margin	25%	
					Total Current Required on Unused Rail	0.00	mA
Regulator	None	N/A	N/A	1	0.00	0.00	mA
					Total Remaining Current Available on Unused Rail	0.00	mA

External Power Source 1	**Component Name**	**Part Number**	**Supply Voltage Range**	**Output Voltage**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
Power Source 1 Selection	12 V Wall Supply	GST60A12-P1J	100-240 VAC	12 V	5,000.00	5,000.00	mA
Power Rails Connected to External Power Source 1	+12V Motor Rail	Direct from 12 V supply	12 V	1	7,007.00	7,007.00	mA
	+5V Linear Regulator	L7805CV	7-35 V input	1	1,500.00	1,500.00	mA
	Unused Rail	N/A	N/A	1	0.00	0.00	mA
					Total Remaining Current Available on External Power Source 1	-3,507.00	mA

External Power Source 2 (Not Used)	**Component Name**	**Part Number**	**Supply Voltage Range**	**Output Voltage**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
Power Source 2 Selection	None	N/A	N/A	N/A	0.00	0.00	mA
Power Rails Connected to External Power Source 2	None	N/A	N/A	1	0.00	0.00	mA
					Total Remaining Current Available on External Power Source 2	0.00	mA

Calculated Battery Life (Not Applicable)							
	Component Name	**Part Number**	**Supply Voltage Range**		**Capacity (mAh)**	**Required By Regulators**	
	No Battery Used	N/A	N/A		0.00	0.00	
					Battery Life	0.00	hours

Notes:

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).

If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit